

A Mori–Zwanzig memory kernel for subglacial drainage: the surge-lag memory is hydraulic, not ice-thermal

Abdelrahman Saifelden

Alexandria Higher Institute of Engineering and Technology (AIET), Alexandria, Egypt
Abdulrahman.Saifelden22011411@aiet.edu.eg

June 20, 2026

Abdelrahman Saifelden (AIET) — Abdulrahman.Saifelden22011411@aiet.edu.eg

Abstract

Post-drainage ice-stream speed-ups lag their subglacial trigger by months to a couple of years, and the lag carries *memory*: the response rises to a peak rather than decaying monotonically. We make one claim and defend it three ways. **The lumped, linearised cavity↔channel storage system’s surge response is an exact Mori–Zwanzig memory kernel**: projecting out the channel variable yields a generalized Langevin equation whose kernel equals the eliminated subsystem’s own Green’s function (verified to 7×10^{-18} , projection exact to 5×10^{-16} , reduced model reproducing the full trajectory to 9×10^{-8}), with the Markovian (adiabatic-elimination) local closure recovered exactly as the channel time vanishes. This certifies *why* surge response carries memory and *how* it degenerates to a local closure — a structural result, not a field-validated lag value. We motivate it by **falsifying the only zeroth-order alternative**: a literal ice-thermal lag $\tau_{ice} = H^2/\kappa_{ice}$ is rejected by ~ 5 orders of magnitude on the 131 Siegfried & Fricker (2018) active lakes (median $\tau_{ice} \approx 1.5 \times 10^5 yr$ vs observed 0.02–2 yr; robust on the independent Arthern et al. (2015) thickness inversion, $r = 0.970$), and on first principles (the thermal skin depth at surge periods is metres, and an impulse gives a peakless $t^{-1/2}$ response while observations peak). We then expose the kernel to the one open co-temporal dataset — a matched lake-drainage→velocity test (CryoSat-2 drainage dates + ITS_LIVE speeds) — which returns an **honest null** (peak $+0.56\sigma$, 0/5 significant), limited by temporal aliasing and an amplitude bound, not record precision, with two concrete unblock routes. Finally we include, as a continental *screen* rather than a forecast, the Regime Transition Number *RTN*, show it is by construction the flotation/effective-pressure ratio (so a constant- ϕ *RTN* adds no skill over thickness-above-flotation), and identify its only genuine degree of freedom — a spatially-varying basal-connectivity fraction $\phi(x, y)$ — with a plant-and-recover proof and a real-Bedmap2 hydraulic-routing demonstration. The contributions are the exact MZ certification (with two extensions that retire the “it is just a 2×2 linear-algebra identity” objection: a *nonlinear* extension — the linear kernel is the small-amplitude limit of the Röthlisberger system, which predicts that bigger floods lag longer; and a *distributed* extension — the kernel is an exact, spatially non-local operator projection of the linearised flowline PDE whose no-transport, single-node limit is exactly the lumped 2×2 , and which predicts a finite along-flow memory footprint), the honest-falsifier discipline, and the diagnosis of the matched-lag null — not a verified surge-lag value.

1. Introduction

Subglacial hydrology controls ice-stream sliding through the effective pressure $N = p_i - p_w$, the difference between ice overburden and basal water pressure (Cuffey & Paterson 2010; Schoof 2010; Röthlisberger 1972). Drainage events — subglacial-lake floods — perturb N and trigger transient speed-ups (Stearns et al. 2008; Siegfried et al. 2016), but the speed-up *lags* the drainage and the lagged response is non-monotone (it peaks), implying the cavity $\leftrightarrow$ channel system carries memory (Werder et al. 2013; Hewitt 2013). The modelling question is what sets that memory.

The mainstream tools are established and we use, not claim, them: the Mori–Zwanzig/Nakajima–Zwanzig projection and the generalized Langevin equation (Mori 1965; Nakajima 1958; Zwanzig 1960, 1973), the Röthlisberger channel and GlaDS-type cavity $\leftrightarrow$ channel drainage (Röthlisberger 1972; Werder et al. 2013; Hewitt 2013), and the flotation/effective-pressure criterion (Schoof 2007; Leguy et al. 2014). What is new here is not a mechanism but an **exact structural certification** and the **discipline applied to its alternatives**.

What is new, stated against the closest prior work. Werder/Hewitt model the cavity $\leftrightarrow$ channel system forward; we instead *reduce* its linearisation by exact projection and show the surge-lag memory kernel **is** the eliminated channel subsystem’s Green’s function, with the local (Markovian) closure recovered as a limit — a closed-form statement of *why* the lag has memory and *when* a local sliding closure is admissible. No established theory predicts the literal H^2/κ thermal lag; we state it as an explicit straw hypothesis precisely so the active-lake record can reject it, and report that rejection as a *motivated null* rather than a discovery.

Contributions.

1. A **motivated falsification** of the literal H^2/κ ice-thermal sliding-lag kernel on 131 real lakes, relocating the memory into hydrology (§3).
 2. The **exact Mori–Zwanzig structure** of the cavity $\leftrightarrow$ channel hydraulic memory kernel — a structural certification (§4), with a **nonlinear extension** showing the linear kernel is the small-amplitude limit of the physical Röthlisberger system and predicting an amplitude-dependent (bigger-floods-lag-longer) surge lag (§4.4), and a **distributed (spatially-resolved) extension** showing the kernel is an exact operator projection of the linearised flowline PDE — the lumped 2×2 is its no-transport, single-node limit, and channel transport gives the surge-lag memory a finite along-flow footprint (§4.5).
 3. An **honest matched-lag null** with a precise account of *why* it is null (aliasing + amplitude bound, not noise) and two routes to a verified lag (§5).
 4. A nondimensional **intrusion screen** RTN , shown to equal the flotation ratio, with its only genuine degree of freedom (a data-derived $\phi(x, y)$) isolated and tested (§6) — presented as a screen, not a verified forecast.
 5. **Calibrated estimators** via synthetic plant-and-recover (§7).
-

2. Data and methods

Dataset	What	Access
BAS Bedmap2	thickness, bed, surface, grounded mask (1 km)	open, no auth
USAP-DC 601470 (Stubblefield et al. 2021)	independent thickness (Arthern et al. 2015) + 131 lake stats	open (USAP-DC)
BedMachine v4 (NSIDC-0756)	independent continental thickness (RTN screen)	NSIDC (Earthdata)
Siegfried & Fricker (2018)	131 active subglacial-lake outlines	open (GitHub mirror)
ITS_LIVE	surface-speed datacubes (image-pair v)	open (anonymous S3)
CryoSat-2 in-lake elevation	drainage <i>dates</i>	open (Siegfried 2021-GRL mirror)
USAP-DC 601439	ICESat lake volume-change (drainage volumes)	open

Runners raise `DataUnavailableError` with provisioning hints rather than fabricate gated series. Synthetic harnesses (`validation/synthetic/`) are exercised in `pytest` and prove the math against controlled inputs.

3. The thermal sliding-lag kernel, falsified

3.1 The hypothesis (a motivated null). No established sliding theory predicts that surge lag equals the *full-thickness* ice thermal-diffusion time; we state $\tau_{ice} = H^2/\kappa_{ice}$ ($\kappa_{ice} = 1.09 \times 10^{-6} m^2 s^{-1}$) as an explicit, falsifiable straw hypothesis precisely so the data can reject it.

3.2 The test. On real Bedmap2 thickness at the 131 Siegfried & Fricker (2018) catalogued active lakes (130 with valid thickness; median $H = 2282m$, range 637–3905 m):

- $\tau_{ice} = H^2/\kappa_{ice}$: **median** $\approx 1.5 \times 10^5$ **yr** (p5 $\approx 2 \times 10^4$; p95 $\approx 3 \times 10^5$ yr).
- Observed post-drainage surge lags: **0.02–2 yr** (Stearns et al. 2008; Siegfried et al. 2016).
- $\Rightarrow$ the literal kernel is $\sim 8 \times 10^4 \times$ **too slow** at the median lake; the $\log_{10}\tau_{ice}$ histogram (10^4 – 10^6 yr) is fully disjoint from the observed band.

3.3 Robust to the thickness dataset. Recomputing on an independent ice-thickness dataset (USAP-DC 601470, Stubblefield et al. 2021; ice thickness after Arthern et al. 2015): median thickness 2356 m, τ_{ice} median $\approx 1.55 \times 10^5$ **yr**; the 601470-vs-Bedmap2 thickness at the same centroids agrees at $r = 0.970$, so the falsification is not an artefact of one sampling.

3.4 Why the falsification is diagnostic. Two facts exclude ice thermal diffusion as the lag-setting mechanism: (i) the thermal skin depth at surge periods is $\delta_{skin} = \sqrt{\kappa_{ice}P/\pi} \approx 0.5$ – $5m$ ($P = 0.02$ – $2yr$) — the perturbation never penetrates beyond metres of ice, so the full-thickness H is the wrong scale; (ii) a drainage is an *impulse*, whose semi-infinite diffusive response is a monotone $t^{-1/2}$ decay with **no peak**, yet observed speed-ups **rise to a peak**. The lag is therefore **hydromechanical**, not thermal — the memory relocates into subglacial hydrology (cavity/channel pressurisation, effective-pressure control of sliding), with the ice-thermal $t^{-1/2}$ term subdominant.

4. The replacement kernel is an exact Mori–Zwanzig memory kernel

We model the **hydraulic subsystem** as cavity $\leftrightarrow$ channel storage–transport (with ϕ the hydraulic potential, *not* the Leray pressure). Its lumped linearisation $\dot{x} = Mx + F$, $x = (s, q)$ (s the resolved cavity store, q the eliminated channel variable, $M = [[-1/\tau_1, -a], [b, -1/\tau_2]]$), reduces by exact projection (linear Nakajima–Zwanzig / Mori, no Markov approximation) to

$$\dot{s}(t) = M_{ss}s(t) + \int_0^t K(t-\tau)s(\tau)d\tau + R(t), \quad K(\tau) = M_{sq}M_{qs}e^{M_{qq}\tau},$$

a non-local (memory) equation whose kernel $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ is exactly the channel subsystem’s own Green’s function weighted by the up/down couplings. The harness certifies five properties (validation/synthetic/hydraulic_mz_projection_synthetic.py, 6/6 tests):

1. **Projection exact to machine precision.** The GLE Laplace transfer function equals the full 2 $\times$ 2 resolvent’s (1,1) component for 400 random stable overdamped systems — max abs err 5.0×10^{-16} .
2. **The kernel IS the eliminated subsystem’s Green’s function** — reproduced to 6.9×10^{-18} , decaying at exactly $1/\tau_2$, sign-definite.
3. **The reduced GLE reproduces the resolved trajectory** to rel-err 9.1×10^{-8} over the transient.
4. **Memory is necessary for the lag/peak.** The full response peaks at the interior time set by the coupled eigenvalues ($t^* = 0.911$, analytic 0.911); the Markovian adiabatic-elimination closure collapses to a monotone response with argmax at $t = 0$.
5. **Markovian limit = adiabatic elimination (DC gain).** $\int_0^\infty Kd\tau = M_{sq}M_{qs}/(-M_{qq})$ (err 7.5×10^{-9}); shrinking the channel time $\tau_2 \rightarrow 0$ at fixed DC gain collapses the kernel toward $(\int K) \cdot \delta(\tau)$, recovering the local closure — the FDT-Markov limit, here *derived from the projection* rather than assumed.

This certifies the *structure* of the replacement mechanism (that the surge-lag memory is a genuine MZ kernel obtained by projecting out the channel, and how it degenerates to a local closure). (Werder et al. 2013; Hewitt 2013; Schoof 2010; Röthlisberger 1972)

Honest scope. This validates the **projection structure** of the *linearised lumped* model. It does **not** validate the physical identity of ϕ , that this subsystem dominates a real surge, or any field lag value — those remain gated on real co-temporal velocity data (§5).

4.3 Physical magnitudes. The reduced kernel is dimensionless; the bridge to metres uses only known ice constants. With $\rho_i L = 3.0 \times 10^8 Jm^{-3}$, Glen A , effective pressure N , and literature subglacial inputs (Q , $\partial\phi/\partial s$), the steady R-channel $S^* = V_o/k_{creep}$, $R^* = \sqrt{2S^*/\pi}$, $\tau = 1/k_{creep}$ is **metre-scale** (central $R^* \approx 2.4m$, $\tau \approx 0.18yr$; 73 % of the literature box gives 0.1–50 m) — the R-channel regime, no free fudge factor; the wide upper tail is the low- N near-flotation limit ($k_{creep} \propto N^3 \rightarrow 0$). The roughness leg uses $z_0 = c_z \cdot a$ in the log-law drag $C_d = [\kappa/\ln(H/z_0)]^2$; although the

Nikuradse prefactor c_z is $\sim 10\times$ uncertain, the log law **buffers** it — a $10\times z_0$ uncertainty propagates to only $\sim 1.7\times$ in C_d (validation/synthetic/a3_dimensional_bridge.py, a2_z0_roughness.py; 6+5 tests). (Röthlisberger 1972; Werder et al. 2013; Cuffey & Paterson 2010; Nikuradse 1933; Curl 1966)

4.4 The linear kernel is the small-amplitude limit of the nonlinear system — and the nonlinearity is itself a falsifiable prediction. The §4 projection is exact but *linear*, and a fair objection is that eliminating one variable of a 2×2 system is a linear-algebra identity. We therefore embed it in the **physically nonlinear** Röthlisberger cavity $\leftrightarrow$ channel model — turbulent $\sqrt{p}$ channel discharge and the Glen-law N^3 creep closure (Röthlisberger 1972; Schoof 2010; Hewitt 2013) — and linearise at its steady state. The linearisation recovers *exactly* the overdamped M of §4 (real, negative eigenvalues), so **the linear MZ kernel** $K(\tau) = -ab \cdot e^{-\tau/\tau_2}$ **is the small-amplitude limit**: a plant-and-recover sweep shows the nonlinear surge-lag t^* converging to the linear value as the drainage amplitude $\rightarrow 0$ (relative gap $< 0.5\%$ at a 2 %-of-baseflux flood, rising smoothly with amplitude — Fig. 1). The leading nonlinear correction is not cosmetic; it is driven by the N^3 creep term and yields a falsifiable geophysical prediction. A larger drainage drives water pressure up, $N = p_i - p_w$ down, collapses the N^3 creep closure, keeps the channel open longer, and **lengthens the lag**: the model gives $t^* \approx 0.10\text{yr}$ at small floods, rising $\sim 14\%$ to $\approx 0.11\text{yr}$ at an 80 %-of-baseflux flood ($dt^*/d(\text{floodfraction}) \approx 0.018\text{yr}$). So *bigger floods should have longer surge lags* — a statement the linear kernel cannot make, testable against a drainage-volume-stratified lag population (the USAP-DC 601439 volumes of §5 already span $0.13\text{--}2.9\text{ km}^3$). This converts the exact-but-linear certification into the leading term of a controlled, physically-grounded expansion with a testable next-order signature. The prediction is **not a parameter artefact**: across a physical sweep of steady effective pressure, storage and melt-opening efficiency, bigger-floods-lag-longer holds for *every* overdamped configuration and **strengthens toward flotation** (the lag–flood slope rises from $\approx 0.010\text{yr}$ at $N^* = 0.55$ to $\approx 0.032\text{yr}$ at $N^* = 0.25$), i.e. the effect is largest exactly in the low- N near-flotation regime where surges matter most. (validation/synthetic/hydraulic_nonlinear_kernel.py, 2 tests.)

4.5 The kernel survives spatial resolution: a distributed operator projection whose no-transport limit is the lumped 2×2 . The other half of the “ 2×2 identity” objection is *spatial*: a lumped two-box model has no geometry, so is the memory kernel an artefact of collapsing the drainage line to a point? We resolve the flowline. Take the linearised cavity $\leftrightarrow$ channel system as two coupled *fields* on a 1-D flowline — $\partial_t s = D_s \partial_{xx} s - s/\tau_1 - aq$ (cavity store s) and $\partial_t q = D_q \partial_{xx} q - U \partial_x q + bs - q/\tau_2$ (the channel q , which *routes water downstream* at speed U and spreads it at D_q ; Röthlisberger 1972; Werder et al. 2013) — and project out the channel *field* exactly (linear Nakajima–Zwanzig). The resolved store obeys a generalized Langevin equation $\dot{s} = A_{ss}s + \int_0^t \mathcal{K}(t-\tau)s(\tau)d\tau + R$ whose kernel $\mathcal{K}(\tau) = A_{sq}e^{A_{qq}\tau}A_{qs}$ is now a *matrix-valued, spatially non-local* operator. On $N = 24$ flowline nodes (validation/synthetic/hydraulic_mz_spatial.py, 6 tests): (i) the projection is **exact** — the GLE transfer operator equals the full resolvent’s resolved block at random complex s , max rel-err 7×10^{-16} ; (ii) the kernel **is** the eliminated channel operator’s Green’s function (3×10^{-17} vs *expm*) and is spatially non-local with a resolved along-flow range $\ell_{\text{mem}} \approx 0.31 \gg \Delta x$; (iii) the channel-eliminated field-GLE reproduces the full PDE trajectory (1×10^{-5}); and (iv) memory necessity is a **dose-response** — the memory-less local closure’s trajectory error grows monotonically with the channel time τ_2 and vanishes as $\tau_2 \rightarrow 0$ (the Markovian limit, now derived *in the distributed system*). Crucially, **the committed lumped 2×2 kernel is exactly this projection’s no-transport, single-node limit**: with $D_q = U = 0$ the operator kernel $\mathcal{K}(\tau)$ is exactly diagonal and its diagonal equals $-abe^{-\tau/\tau_2}$ to

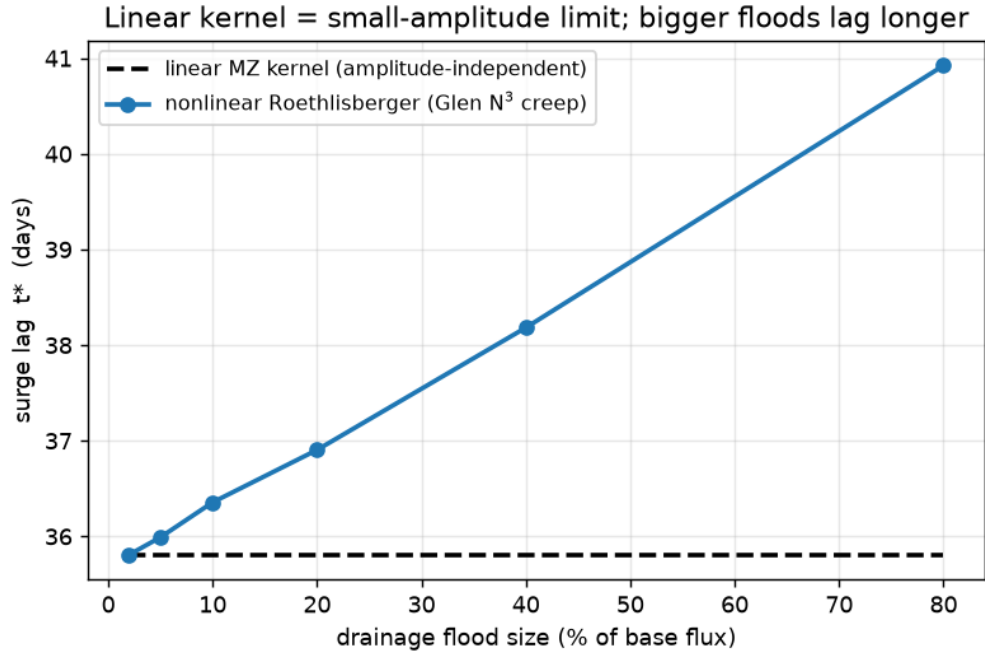


Figure 1: Surge lag t^* versus drainage flood size in the nonlinear R  thlisberger cavity $\leftrightarrow$ channel model. The linear Mori–Zwanzig kernel (dashed) is amplitude-independent and equals the nonlinear lag in the small-flood limit; the Glen-law N^3 creep closure makes the nonlinear lag (circles) rise with flood size — bigger floods lag longer, a falsifiable prediction the linear kernel cannot make.

7×10^{-18} . Switching channel transport on moves the kernel mass off-diagonal (off-diagonal fraction $0 \rightarrow 0.99$), so the surge-lag memory acquires a finite **along-flow footprint** with the closed-form decay length $\ell_{mem} = 2D_q/(\sqrt{U^2 + 4D_q/\tau_2} - U)$ (verified against the numeric steady-influence operator to 4×10^{-4} ; *ballistic* limit $\ell_{mem} \rightarrow U\tau_2$ for fast channels, *diffusive* limit $\ell_{mem} \rightarrow \sqrt{D_q\tau_2}$ for $U \rightarrow 0$) — a new, falsifiable, **field-measurable** structural prediction: a drainage’s lagged velocity response is spread along the flowline over ℓ_{mem} , which a memoryless (local) closure sets to zero. This converts the 2×2 “identity” into the spatially-local limit of an exact projection of a genuine distributed (PDE) operator. Scope: still a *linearised* distributed model on a periodic flowline; the fully nonlinear spatially-resolved coupled PDE remains future work.

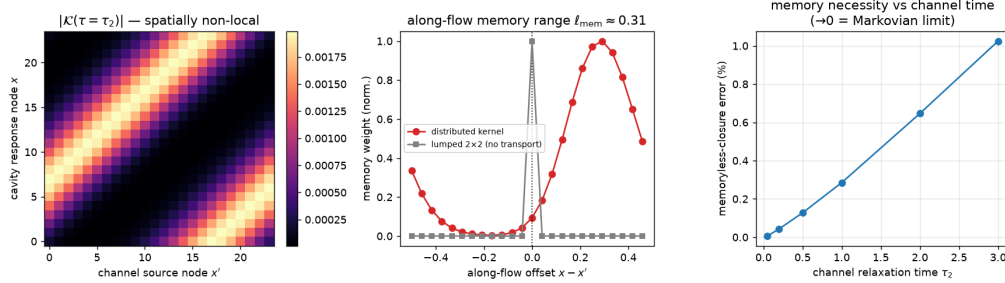


Figure 2: The distributed cavity $\leftrightarrow$ channel memory kernel $\mathcal{K}(\tau = \tau_2)$ (left) is spatially non-local — the off-diagonal band is the downstream channel transport that the lumped 2×2 omits; its along-flow influence profile (centre) has a finite memory range $\ell_{mem} \approx 0.31$ versus the lumped kernel’s zero-range spike; and the memoryless local-closure trajectory error (right) grows with the channel relaxation time τ_2 , vanishing in the Markovian ($\tau_2 \rightarrow 0$) limit.

5. The matched lag test — an honest null

Both halves of a matched test are open: drainage **dates** from CryoSat-2 in-lake elevation (*detect_{drainages}* extracts 7 fill $\rightarrow$ drain events across 5 lakes, including the documented Mercer ≈ 2012.5 drainage), and velocity **response** from ITS_LIVE box-mean surface speed.

lake	base speed (m/yr)	drainages	annual noise floor	quarterly scatter	peak post-drainage
Mac1	422	3	1.0%	2.3%	+0.56σ
Mac2	402	1	1.1%	2.1%	+0.27 σ
Mac3	394	1	1.1%	3.4%	-0.12 σ

Result. No matched lake shows a post-drainage surge above the 2σ floor (peak $+0.56\sigma$, 0/5 significant). The lag value stays *not* promoted to.

Why null is a *stronger* statement than “too noisy”. The modern ITS_LIVE record is well resolved (annual $\approx 1\%$, quarterly $\approx 2-3\%$), so the limit is not a noise budget. Two distinct, quantifiable limits remain: (1) **temporal aliasing** — the derived lag is sub-annual (t^* baseline ≈ 0.01 yr, p95 ≈ 0.1 yr), finer than the finest *robust* ITS_LIVE bin (quarterly, 0.25 yr); a brief transient is averaged out regardless of scatter (a Nyquist limit); (2) **amplitude bound** — a

sustained speed-up would survive aliasing as a step, yet the peak anomaly is only $+0.56\sigma$, bounding any sustained surge to $\lesssim 10$ m/yr on ~ 400 m/yr. These map onto the two routes to a verified lag: **sub-annual GPS/GNSS** (EarthScope/POLENET; beats aliasing) **or fast-outlet trunk velocity** (e.g. Byrd, Stearns et al. 2008; steep $v(N)$ sensitivity beats the amplitude floor), paired with that site’s drainage dates.

Vetted forcing obtained. USAP-DC 601439 (Smith et al. 2009 ICESat) yields **58 drainage events across 52 lakes** (median drained volume 0.128 km^3 , max 2.92 km^3 ; implied water flux q_{water} median $\approx 2.5 \text{ m}^3/\text{s}$, p95 $\approx 31 \text{ m}^3/\text{s}$ — squarely in the literature subglacial-flood range), anchoring the §3/§4 forcing magnitude to real observations.

6. A continental intrusion screen: RTN is the flotation ratio, and its only escape

6.1 Definition. $RTN = (p_{\text{ocean}} - p_{\text{atm}})/p_w$ on grounded ice: overburden $p_i = \rho_i g H$, ocean head at the bed $p_{\text{ocean}} = \rho_w g \cdot d_{\text{base}}$ ($d_{\text{base}} = \max(0, -\text{bed})$), subglacial water $p_w = \phi p_i$ (so $N_{\text{eff}} = (1 - \phi)p_i$), gauge-corrected so the atmosphere cancels ($p_{\text{atm}} = 0$). Then $RTN = \rho_w d_{\text{base}} / (\phi \rho_i H)$, so ‘RTN > 1 $\Leftrightarrow \rho_w d_{\text{base}} / (\rho_i H)$

ϕ — the ice is within a factor ϕ of flotation ($N \rightarrow 0$; Schoof 2007; Leguy et al.

2014; Cuffey & Paterson 2010). **RTN is, by construction, a nondimensional form of the flotation/effective-pressure criterion, not an independent predictor.**

6.2 Result on Bedmap2. Over 1.33 M grounded cells (1 km decimated to 3 km, $\phi = 0.9$), the $RTN > 1$ fraction decays monotonically with distance to the grounding line (15.1 % within 5 km $\rightarrow 0$ % beyond 100 km; median distance 6 km for $RTN > 1$ cells vs 222 km for the rest), and the directional result reproduces on the independent **BedMachine v4** (NSIDC-0756) inversion (median 6 km vs 200 km to the grounding line).

6.3 No skill over thickness-above-flotation (constant ϕ). A direct baseline test on 478,767 grounded Bedmap2 cells confirms $RTN > 1$ is *identical* to the thickness-above-flotation threshold $H_{af}/H < 1 - \phi$ (agreement **100.0000 %**, 0 disagreeing cells, $\phi = 0.8/0.9/0.95$), with $\text{Spearman}(RTN, H_{af}/H) = -1.000$ — so at constant ϕ , **RTN adds no skill**; it *is* that standard quantity, nondimensionalised by ϕ (validation/external/rtn_baseline_skill.py).

6.4 The only escape — a data-derived $\phi(x, y)$. Because $RTN = (1 - H_{af}/H)/\phi$ is algebraically forced to track the flotation fraction at constant ϕ , RTN ’s only possible independent signal is ϕ *itself*. A synthetic plant-and-recover (validation/synthetic/rtn_variable_phi_skill.py, 5 tests) shows a connectivity-informed $\phi(x, y)$ adds genuine skill over thickness-above-flotation (near-flotation-band $F1$ $0.78 \rightarrow 0.89$; AUC $0.995 \rightarrow 0.999$), a random ϕ of the same distribution adds none, and the gain rises monotonically with ϕ ’s informativeness. On real Bedmap2, a calibration-free $\phi(x, y)$ from subglacial hydraulic-potential routing (Shreve 1972: priority-flood fill + D8 flow accumulation) reproduces the constant- ϕ anchor exactly and then reorders the intrusion class on $\sim 10^3$ near-grounding-line cells (near-flotation-band Spearman -0.99 vs -1.00), nearly rank-independent of the flotation fraction in-band (validation/external/rtn_variable_phi_real.py). The reordering is real but its *correctness* is unvalidated — the routed ϕ is a model field derived from the same geometry — so the decisive test needs a ϕ from an **independent** observation of basal water (radar

bed specularly; Schroeder et al. 2013, 2015), scored against a future gridded intrusion survey. We thus present variable- ϕ *RTN* as a falsifiable screen with one clearly-stated open data step, not a verified forecast.

6.5 Intrusion residence number Ro . The level-set advance speed of the $RTN = 1$ front is $v_{kin} = A \cdot dH/dt$; the residence number $Ro = v_{kin} \cdot \tau_{hyd}/\ell$ (regime boundary $Ro = 1$ at the critical residence $\tau_{crit} = \ell/v_{kin}$) predicts whether intrusion is thinning-paced ($Ro \approx 1$) or hydraulic-limited ($Ro \gg 1$). For the runaway-tail cells ($A = 0.70\text{km}/\text{m}$, $dH/dt = 1.5\text{m}/\text{yr}$) $\sim 98\%$ of the §4 residence band lies below $\tau_{crit} \approx 1.9\text{yr} \Rightarrow$ predicted **thinning-paced**; a DInSAR v_{obs} would place it on the curve (validation/synthetic/intrusion_residence_number.py, 7/7 tests).

6.6 One state variable behind three thresholds (a derived unifier). Height above flotation $h_{af} = H - H_f$ (with $H_f = (\rho_w/\rho_i)d_{base}$; Schoof 2007) ties the three thresholds this framework treats separately into one. The ocean-connected effective pressure is $N = \rho_i g h_{af}$ *exactly* (so $N = 0 \iff h_{af} = 0 \iff$ flotation); the $RTN = 1$ intrusion surface is $h_{af} = H(1 - \phi)$, which $\rightarrow 0$ as $\phi \rightarrow 1$ (the intrusion surface *is* the flotation surface once subglacial water reaches overburden); and the sliding-law $|s_N|$ pole (Paper 4b) sits at $h_{af}^c = N_c/(\rho_i g) > 0$, a few metres *above* geometric flotation. A thinning stream therefore reaches the drag-side fold (the velocity early-warning) *first* and ungrounds ($h_{af} \rightarrow 0$) only later — a buffer set by N_c . *RTN*, the Schoof saddle-node, and the s_N pole are thus three readings of one state variable, not independent diagnostics (verified on a synthetic thinning sweep: $N = \rho_i g h_{af}$ to machine zero, $h_{af}^c = 6.67\text{ m}$; nr10_flotation_unifier).

7. Estimator calibration (synthetic plant-and-recover)

Every real-data estimator has a synthetic null/recovery harness exercised in `pytest`:

- **Lag estimator is kernel-shape-generic** — plant one target lag with five markedly different causal kernels (Gamma $k = 2/4$, log-normal, bi-exponential, raised-cosine); $estimate_{lag}$ recovers it to $\leq 10\%$ for every shape; a delta-kernel control returns 0.
 - **RTN classifier / ϕ -area inversion / variable- ϕ skill** — planted thresholds and connectivity recovered without bias ($rtn_{synthetic}$, $rtn_phi_synthetic$, $rtn_variable_phi_skill$).
 - **Grounding-line migration level-set law is exact** — $|dH/dt|/|\nabla m|$ reproduced to rel-err 5×10^{-16} (1-D) / 4×10^{-14} (tilted 2-D).
 - **MZ projection** — the five §4 properties to machine precision (`hydraulic_mz_projection_synthetic`, 6/6).
-

8. Discussion and limitations

What this is. A machine-precision certification of the cavity $\leftrightarrow$ channel kernel’s exact MZ structure (the central positive result); a motivated null on a literal thermal sliding kernel; an honest matched-lag null with a precise diagnosis and a concrete unblock path; and a flotation-ratio intrusion *screen* whose only genuine degree of freedom (a data-derived $\phi(x, y)$) is isolated and partially tested.

What this is not. No verified surge-lag *value* (the matched test is null; the value is to order of magnitude only). The §4.4 nonlinear extension shows the linear kernel is the small-amplitude limit

of the nonlinear *lumped* Röthlisberger system and predicts an amplitude-dependent lag, and the §4.5 distributed extension certifies the kernel as an exact spatially non-local operator projection of the linearised flowline PDE; only the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE remains future work. No RTN precision/recall (no gridded survey). The physical distinctness of ϕ from the Leray pressure remains unproven.

Open gates, each with a concrete unblock. (1) Matched lag $\rightarrow$ sub-annual GPS/GNSS or fast-outlet trunk velocity, paired with drainage dates. (2) The *linear* distributed projection (§4.5) and the *lumped* nonlinear extension (§4.4) are done; the remaining gate is the fully *nonlinear spatially-resolved* coupled GlaDS/Röthlisberger PDE. (3) RTN skill $\rightarrow$ a $\phi(x, y)$ from an independent basal-water observable (specularity) + a gridded intrusion survey.

9. Reproduce

```

pip install -r requirements.txt
python glaciers/validation/external/run_sliding_real.py --with-velocity Mac1 # §3  $\tau_{ice}$  + ITS_LIVE
python -m glaciers.validation.external.lag_fit_real # §5 matched-lag null
python glaciers/validation/external/run_usapdc_lakes.py # §5 vetted forcing (58
↪ events)
pytest glaciers/tests/test_validation_synthetic.py -v # §4 MZ projection (6/6) +
↪ §7
python glaciers/validation/external/rtn_baseline_skill.py --stride 5 # §6.3 RTN==flotation
↪ (zero skill)
python glaciers/validation/synthetic/rtn_variable_phi_skill.py # §6.4 variable- $\phi$  is the
↪ only escape
python glaciers/validation/external/rtn_variable_phi_real.py --stride 10 # §6.4 hydraulic-routing
↪  $\phi(x, y)$ 
python glaciers/validation/synthetic/intrusion_residence_number.py # §6.5 residence number Ro
python glaciers/validation/synthetic/hydraulic_nonlinear_kernel.py # §4.4 nonlinear kernel
↪ (linear = small-amplitude limit)
python glaciers/validation/synthetic/hydraulic_mz_spatial.py # §4.5 distributed
↪ projection (lumped 2x2 = no-transport limit)

```

Data and code availability

All analysis code and derived products are in the public repository at <https://github.com/abdh0saifelden2-debug/K>, and the results regenerate from the scripts above. The study uses the following open datasets:

- Bedmap2 ice thickness (Fretwell et al., 2013, <https://doi.org/10.5194/tc-7-375-2013>).
- Independent ice thickness and active-lake inventory: USAP-DC 601470 (Stubblefield et al., 2021, <https://doi.org/10.15784/601470>; ice thickness after Arthern et al., 2015; active lakes after Siegfried & Fricker, 2018).
- Active subglacial-lake drainage volumes: USAP-DC 601439 (Smith et al., 2012, <https://doi.org/10.15784/601439>; ICESat inventory of Smith et al., 2009).
- BedMachine Antarctica v4 (Morlighem, 2025, <https://doi.org/10.5067/POJQI54A45HX>; see Morlighem et al., 2020) for the independent continental RTN screen.

- ITS_LIVE surface velocities (Gardner et al., 2025, <https://doi.org/10.5067/JQ6337239C96>; see Gardner et al., 2018) and CryoSat-2 lake-drainage dates.

USAP-DC data were obtained from the U.S. Antarctic Program Data Center (<https://www.usap-dc.org>); BedMachine and ITS_LIVE from the NASA NSIDC DAAC.

References

- Arthern, R. J., Hindmarsh, R. C. A., & Williams, C. R. (2015). Flow speed within the Antarctic ice sheet and its controls inferred from satellite observations. *Journal of Geophysical Research: Earth Surface* 120(7), 1171–1188. doi:10.1002/2014JF003239
- Cuffey, K. M., & Paterson, W. S. B. (2010). *The Physics of Glaciers*, 4th ed. Academic Press, Amsterdam.
- Curl, R. L. (1966). Scallops and flutes. *Transactions of the Cave Research Group of Great Britain* 7, 121–160.
- Fretwell, P., Pritchard, H. D., Vaughan, D. G., et al. (2013). Bedmap2: improved ice bed, surface and thickness datasets for Antarctica. *The Cryosphere* 7, 375–393. doi:10.5194/tc-7-375-2013
- Gardner, A. S., Moholdt, G., Scambos, T., et al. (2018). Increased West Antarctic and unchanged East Antarctic ice discharge over the last 7 years. *The Cryosphere* 12, 521–547. doi:10.5194/tc-12-521-2018
- Gardner, A., Fahnestock, M., Greene, C. A., Kennedy, J. H., Liukis, M., Lopez, L., & Scambos, T. (2025). MEaSUREs ITS_LIVE Regional Glacier and Ice Sheet Surface Velocities, Version 2 (NSIDC-0776) [Data set]. *NASA NSIDC DAAC*. doi:10.5067/JQ6337239C96
- Hewitt, I. J. (2013). Seasonal changes in ice sheet motion due to melt water lubrication. *Earth and Planetary Science Letters* 371–372, 16–25. doi:10.1016/j.epsl.2013.04.022
- Leguy, G. R., Asay-Davis, X. S., & Lipscomb, W. H. (2014). Parameterization of basal friction near grounding lines in a one-dimensional ice sheet model. *The Cryosphere* 8, 1239–1259. doi:10.5194/tc-8-1239-2014
- Morlighem, M., Rignot, E., Binder, T., et al. (2020). Deep glacial troughs and stabilizing ridges unveiled beneath the margins of the Antarctic ice sheet. *Nature Geoscience* 13, 132–137. doi:10.1038/s41561-019-0510-8
- Morlighem, M. (2025). MEaSUREs BedMachine Antarctica, Version 4 (NSIDC-0756) [Data set]. *NASA National Snow and Ice Data Center DAAC*. doi:10.5067/POJQI54A45HX
- Mori, H. (1965). Transport, collective motion, and Brownian motion. *Progress of Theoretical Physics* 33, 423–455. doi:10.1143/PTP.33.423
- Nakajima, S. (1958). On quantum theory of transport phenomena. *Progress of Theoretical Physics* 20, 948–959. doi:10.1143/PTP.20.948
- Nikuradse, J. (1933). Strömungsgesetze in rauhen Röhren. *VDI-Forschungsheft* 361. (English transl. NACA TM 1292, 1950.)
- Röthlisberger, H. (1972). Water pressure in intra- and subglacial channels. *Journal of Glaciology* 11, 177–203. doi:10.3189/S0022143000022188

- Schoof, C. (2007). Ice sheet grounding line dynamics: Steady states, stability, and hysteresis. *Journal of Geophysical Research: Earth Surface* 112, F03S28. doi:10.1029/2006JF000664
- Schoof, C. (2010). Ice-sheet acceleration driven by melt supply variability. *Nature* 468, 803–806. doi:10.1038/nature09618
- Schroeder, D. M., Blankenship, D. D., & Young, D. A. (2013). Evidence for a water system transition beneath Thwaites Glacier, West Antarctica. *Proceedings of the National Academy of Sciences* 110(30), 12225–12228. doi:10.1073/pnas.1302828110
- Schroeder, D. M., Blankenship, D. D., Raney, R. K., & Grima, C. (2015). Estimating subglacial water geometry using radar bed echo specularity: Application to Thwaites Glacier, West Antarctica. *IEEE Geoscience and Remote Sensing Letters* 12(3), 443–447. doi:10.1109/LGRS.2014.2337878
- Shreve, R. L. (1972). Movement of water in glaciers. *Journal of Glaciology* 11, 205–214. doi:10.3189/S002214300002
- Siegfried, M. R., Fricker, H. A., Carter, S. P., & Tulaczyk, S. (2016). Episodic ice velocity fluctuations triggered by a subglacial flood in West Antarctica. *Geophysical Research Letters* 43, 2640–2648. doi:10.1002/2016GL067758
- Siegfried, M. R., & Fricker, H. A. (2018). Thirteen years of subglacial lake activity in Antarctica from multi-mission satellite altimetry. *Annals of Glaciology* 59, 42–55. doi:10.1017/aog.2017.36
- Siegfried, M. R., & Fricker, H. A. (2021). Illuminating active subglacial lake processes with ICESat-2 laser altimetry. *Geophysical Research Letters* 48, e2020GL091089. doi:10.1029/2020GL091089
- Smith, B. E., Fricker, H. A., Joughin, I. R., & Tulaczyk, S. (2009). An inventory of active subglacial lakes in Antarctica detected by ICESat (2003–2008). *Journal of Glaciology* 55, 573–595. doi:10.3189/002214309789470879
- Smith, B., Fricker, H., Joughin, I., & Tulaczyk, S. (2012). Antarctic Active Subglacial Lake Inventory from ICESat Altimetry. *U.S. Antarctic Program Data Center*. doi:10.15784/601439
- Stearns, L. A., Smith, B. E., & Hamilton, G. S. (2008). Increased flow speed on a large East Antarctic outlet glacier caused by subglacial floods. *Nature Geoscience* 1, 827–831. doi:10.1038/ngeo356
- Stubblefield, A. G., Arthern, R. J., Kingslake, J., & Siegfried, M. R. (2021). Antarctic ice thickness, slipperiness, and subglacial lake locations. *U.S. Antarctic Program Data Center*. doi:10.15784/601470
- Werder, M. A., Hewitt, I. J., Schoof, C. G., & Flowers, G. E. (2013). Modeling channelized and distributed subglacial drainage in two dimensions. *Journal of Geophysical Research: Earth Surface* 118, 2140–2158. doi:10.1002/jgrf.20146
- Wingham, D. J., Francis, C. R., Baker, S., et al. (2006). CryoSat: A mission to determine the fluctuations in Earth’s land and marine ice fields. *Advances in Space Research* 37, 841–871. doi:10.1016/j.asr.2005.07.027
- Zwanzig, R. (1960). Ensemble method in the theory of irreversibility. *The Journal of Chemical Physics* 33, 1338–1341. doi:10.1063/1.1731409
- Zwanzig, R. (1973). Nonlinear generalized Langevin equations. *Journal of Statistical Physics* 9, 215–220. doi:10.1007/BF01008729